

Fifteenth day of class - 10/16

Fun fact:

$$e^{\pi\sqrt{163}} \approx 262537412640768743.99999999999925$$

is almost exactly an integer. Why?

1. Recall that we defined $j(z)$ to be the j -invariant of the elliptic curve with the lattice $L_z = \{a + bz : a, b \in \mathbb{Z}\}$.

If the lattice L_z is closed under multiplication (which makes it a ring), then the elliptic curve with lattice L_z has extra maps to itself (which send $\mathcal{O}$ to $\mathcal{O}$).

2. In this situation, it is possible to scale the lattice L_z so that the elliptic curve

$$E : y^2 = 4x^3 - g_2(L_z)x - g_3(L_z)$$

has coefficients that are algebraic numbers. In this situation, $j(E)$ is the root of a *monic* polynomial with integer coefficients.

3. If the set of numbers in L_z has the unique factorization property, then $j(E) \in \mathbb{Z}$.

4. The “last” time this occurs is when $z = \frac{1+i\sqrt{163}}{2}$. Recall that

$$j(z) = q^{-1} + 744 + 196884q + \dots$$

where $q = e^{2\pi iz} = -e^{-\pi\sqrt{163}}$. So we have $j(z) \in \mathbb{Z}$ and

$$j(z) = -e^{\pi\sqrt{163}} + 744 + 196884(-e^{-\pi\sqrt{163}}) + \dots$$

The term $196884(-e^{-\pi\sqrt{163}}) + \dots$ and later terms are very small. So

$$e^{\pi\sqrt{163}} = -j(z) + 744 + \text{small}$$

is almost exactly an integer.

Mordell's theorem

Our main goal for the next while is to work on proving Mordell's theorem, that $C(\mathbb{Q})$ is a finitely-generated abelian group. This relies on four lemmas.

Recall that if $P = (x, y)$ is a point on $C(\mathbb{Q})$, with $x = \frac{a}{b}$ and $\gcd(a, b) = 1$, we define

$$h(P) = \max\{\log(|a|), \log(|b|)\}.$$

(Here $\log$ is the natural logarithm.)

The four lemmas

Lemma 1. For any real number M , the set $\{P \in C(\mathbb{Q}) : h(P) \leq M\}$ is finite.

Lemma 2. Let P_0 be a fixed point on C . There is a constant κ_0 , depending on P_0 , a , b and c so that

$$h(P + P_0) \leq 2h(P) + \kappa_0 \quad \forall P \in C(\mathbb{Q}).$$

Lemma 3. *There is a constant κ , depending on a , b and c so that*

$$h(2P) \geq 4h(P) - \kappa \quad \forall P \in C(\mathbb{Q}).$$

This means that if you were to write down the x -coordinate of $2P$, the numerator and denominator would roughly have about 4 times as many digits as that of P . (In general, the x -coordinate of mP has about m^2 times as many digits as that of P .)

Definition. *If G is a group and H is a subgroup of G , then index of H in G $[G : H]$ is the number of distinct left cosets of H .*

If G is finite, every left coset of H has size equal to $|H|$ and every element of G is in exactly one left coset. Then,

$$[G : H] = \frac{|G|}{|H|}.$$

However, we will apply this idea with infinite groups also.

Let $2C(\mathbb{Q}) = \{P + P : P \in C(\mathbb{Q})\}$. This is a subgroup of $C(\mathbb{Q})$.

Lemma 4. *The index $[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ is finite.*

Theorem (The Descent Theorem). *Let Γ be an abelian group. Suppose that there is a function $h : \Gamma \rightarrow [0, \infty)$ with the following three properties.*

- (a) *For every real number M , the set $\{P \in \Gamma : h(P) \leq M\}$ is finite.*
- (b) *For every $P_0 \in \Gamma$, there is a constant κ_0 so that*

$$h(P + P_0) \leq 2h(P) + \kappa_0.$$

- (c) *There is a constant κ so that $h(2P) \geq 4h(P) - \kappa$ for all $P \in \Gamma$, and*
- (d) *The subgroup 2Γ has finite index in Γ .*

Then Γ is finitely generated.

Example: Let $C : y^2 = x^3 - 2x$.

1. If $P_0 \in C(\mathbb{Q})$ and $P \in C(\mathbb{Q})$, then

$$h(P + P_0) \leq 2h(P) + 2h(P_0) + 2.5.$$

2. We also have that if $P \in C(\mathbb{Q})$, then

$$h(2P) \geq 4h(P) - 1.4.$$

3. We have that $[E(\mathbb{Q}) : 2E(\mathbb{Q})] = 4$, and coset representatives are $\{(0 : 1 : 0), (0, 0), (2, 2), (-1, 1)\} = \{Q_1, Q_2, Q_3, Q_4\}$.

Now, suppose that

$$P = \left(-\frac{77227592}{3263037129}, \frac{40547346141484}{186394469919867} \right).$$

We have that $P - (0, 0) = 2P_1$, where $P_1 = (338, 6214)$. We have $P_1 - (2, 2) = 2P_2$ where $P_2 = (2, 2) = Q_3$.

Thus, we have $P_1 = Q_3 + 2Q_3 = 3Q_3$. Then $P = 2(3Q_3) + (0, 0) = 6Q_3 + Q_2$.

(At the end, we got down to a representation of P in terms of $\{Q_1, Q_2, Q_3, Q_4\}$. However, if we choose “bad” coset representatives, we might get points at the end with lower height than the Q_i .)

Proof of descent theorem. By (d), $[\Gamma : 2\Gamma] = n$, so there are n cosets of 2Γ in Γ . This means that there are $Q_1, Q_2, \dots, Q_n$ so that

$$\Gamma = \bigcup_{i=1}^n 2\Gamma + Q_i.$$

Another way of saying this is that for any $P \in \Gamma$, there exists Q_i so that $P - Q_i \in 2\Gamma$.

Applying property 2 with $P_0 = -Q_i$ we get that $h(P - Q_i) \leq 2h(P) + \kappa_i$ for some $\kappa_i \in \mathbb{R}$. Define $\kappa' = \max_i \kappa_i$.

We'll show that $\{Q_i\} \cup \{R \in C(\mathbb{Q}) : h(R) \leq \kappa + \kappa'\}$ generate Γ .

Choose a specific $P \in C(\mathbb{Q})$. There exists $Q_{i_1} \in \{Q_1, Q_2, \dots, Q_n\}$ so that

$$P = Q_{i_1} + 2P_1.$$

Now $P_1 \in C(\mathbb{Q})$ and so there exists Q_{i_2} so $P_1 = Q_{i_2} + 2P_2$. We can keep going in this fashion and write

$$\begin{aligned} P - Q_{i_1} &= 2P_1 \\ P_1 - Q_{i_2} &= 2P_2 \\ &\dots \\ P_{m-1} - Q_{i_m} &= 2P_m. \end{aligned}$$

Now, we can rewrite and get

$$\begin{aligned} P &= Q_{i_1} + 2P_1 \\ &= Q_{i_1} + 2Q_{i_2} + 2P_2 \\ &= Q_{i_1} + 2Q_{i_2} + 4Q_{i_3} + 4P_3 \\ &\dots \\ &= Q_{i_1} + 2Q_{i_2} + 4Q_{i_3} + \dots + 2^m Q_{i_m} + 2^m P_m. \end{aligned}$$

We have that $h(P - Q_i) \leq 2h(P) + \kappa'$, where κ' was defined above. Now, we have

$$4h(P_j) \leq h(2P_j) + \kappa = h(P_{j-1} - Q_{i_j}) + \kappa \leq 2h(P_{j-1}) + \kappa' + \kappa.$$

Thus, $h(P_j) \leq \frac{1}{2}h(P_{j-1}) + \frac{\kappa + \kappa'}{4}$. We can rewrite this as

$$h(P_j) \leq \frac{3}{4}h(P_{j-1}) - \frac{1}{4}(h(P_{j-1}) - (\kappa' + \kappa)).$$

From this, we see that if $h(P_{j-1}) \geq \kappa' + \kappa$, then $h(P_j) \leq \frac{3}{4}h(P_{j-1})$. By choosing m large enough, we can arrange for $h(P_m) \leq \kappa' + \kappa$. Thus, we can write

$$P = a_1Q_1 + a_2Q_2 + \cdots + a_nq_n + 2^mR$$

where $h(R) \leq \kappa' + \kappa$, and thus any $P \in C(\mathbb{Q})$ can be generated by $\{Q_i\} \cup \{R \in C(\mathbb{Q}) : h(R) \leq \kappa' + \kappa\}$. $\square$

Claim: If $P = (x, y)$ is a rational point on an elliptic curve $y^2 = x^3 + ax^2 + bx + c$ with $a, b, c \in \mathbb{Z}$, then we can write $x = \frac{m}{e^2}$, $y = \frac{n}{e^3}$ with $m, n, e \in \mathbb{Z}$ and $\gcd(m, e) = \gcd(n, e) = 1$.

Proof: This follows from the material in Section II.4, but here's a reasonably simple direct proof. Write $x = m/M$ and $y = n/N$ with $\gcd(m, M) = \gcd(n, N) = 1$. Plugging this in to the equation gives us

$$\frac{n^2}{N^2} = \frac{m^3 + am^2M + bmM^2 + cM^3}{M^3}.$$

Now, the denominator of the left hand side is equal to N^2 . Since $\gcd(m, M) = 1$ and M divides the last three terms of the numerator, we have that $\gcd(m^3 + am^2M + bmM^2 + cM^3, M) = 1$. This shows that the right hand side is reduced as well, and so its denominator is M^3 .

Hence $N^2 = M^3$. Thus, M^3 and hence M is a perfect square. Let $M = e^2$. Then $N^2 = M^3 = e^6$ and so $N = e^3$. This proves the claim. QED

Proof of Lemma 2. Now, if $x = \frac{m}{e^2}$, then $H(P) = \max\{|m|, e^2\}$ and so $|m| \leq H(P)$ and $e^2 \leq H(P)$.

Claim: There is a constant K , depending on a, b and c so that $|n| \leq KH(P)^{3/2}$.

Proof of Claim: Taking $\left(\frac{n}{e^3}\right)^2 = \left(\frac{m}{e^2}\right)^3 + a\left(\frac{m}{e^2}\right)^2 + b\left(\frac{m}{e^2}\right) + c$ and clearing denominators gives

$$n^2 = m^3 + ae^2m^2 + be^4m + ce^6.$$

This gives

$$\begin{aligned} |n^2| &\leq |m^3| + |ae^2m^2| + |be^4m| + |ce^6| \\ &\leq H(P)^3 + |a|H(P)^3 + |b|H(P)^3 + |c|H(P)^3 \\ &\leq (1 + |a| + |b| + |c|)H(P)^3. \end{aligned}$$

Taking the square root, we get that $n \leq KH(P)^{3/2}$, where

$$K + \sqrt{1 + |a| + |b| + |c|}.$$

QED Claim

Strategy - write out the formula for $P + P_0$ and use the triangle inequality. If $P_0 = \mathcal{O} = (0 : 1 : 0)$, it's easy. (We can take $\kappa_0 = 0$.)

Fix a $P_0 \in C(\mathbb{Q})$ with $P_0 \neq \mathcal{O}$. To prove the statement, we can exclude a finite number of values of P , because if $h(P + P_0) \leq 2h(P) + \kappa_0$ for all P not in S , then we can define

$$\kappa_1 = \max_{P \in S} 2h(P) - h(P + P_0)$$

then $h(P + P_0) \leq 2h(P) + \max\{\kappa_0, \kappa_1\}$.

We will take $S = \{P_0, -P_0, \mathcal{O}\}$, so we can avoid using the duplication formula.

Let $P_0 = (x_0, y_0)$, $P = (x, y)$, and $P + P_0 = (\xi, \eta)$. The addition law gives

$$\xi + x + x_0 = \lambda^2 - a, \quad \lambda = \frac{y - y_0}{x - x_0}.$$

We get

$$\begin{aligned} \xi &= \lambda^2 - a - x - x_0 \\ &= \frac{(y - y_0)^2 - (x - x_0)^2(a + x + x_0)}{(x - x_0)^2}. \end{aligned}$$

When we simplify this, we can use $y^2 = x^3 + ax^2 + bx + c$ and get that $y^2 - x^3 = ax^2 + bx + c$. We will end up with an expression of the form

$$\xi = \frac{Ay + Bx^2 + Cx + D}{Ex^2 + Fx + G}$$

where A, B, C, D, E, F and G are all constants depending only on a, b, c, x_0 and y_0 .

We multiply through by the least common multiple of the denominators of A through G and assume that they are integers therefore.

Now, setting $x = m/e^2$ and $y = n/e^3$ we get

$$\xi = \frac{Ane + Bm^2 + Cme^2 + De^4}{Em^2 + Fme^2 + Ge^4}.$$

Thus,

$$H(P + P_0) = H(\xi) = \max\{|Ane + Bm^2 + Cme^2 + De^4|, |Em^2 + Fme^2 + Ge^4|\}.$$

We use that $e^2 \leq H(P)$, $|m| \leq H(P)$ and $|n| \leq KH(P)^{3/2}$. We get

$$\begin{aligned} |Ane + Bm^2 + Cme^2 + De^4| &\leq (|AK| + |B| + |C| + |D|)H(P)^2 \\ |Em^2 + Fme^2 + Ge^4| &\leq (|E| + |F| + |G|)H(P)^2. \end{aligned}$$

Thus,

$$H(P + P_0) \leq H(\xi) \leq \max\{|AK| + |B| + |C| + |D|, |E| + |F| + |G|\} H(P)^2.$$

Letting $\kappa_0 = \log(\max\{|AK| + |B| + |C| + |D|, |E| + |F| + |G|\})$, we get

$$h(P + P_0) \leq \log h(P + P_0) = 2h(P) + \kappa_0.$$

□